

Time: 60 minutes

Max marks: 20

Instructions:

- Do not plagiarize. Do not assist your classmates in plagiarism.
- Show your full solution for the questions to get full credit.
- Attempt all questions that you can.
- In the unlikely case a question is not clear, discuss it with an invigilating TA. Please ensure that you clearly include any assumptions you make, even after clarification from the invigilator.

1. (7 points) **Deep Learning CNN & loss functions [CO-1]**

State True or False with justification (or a counter-example) as applicable.

- (a) (1 point) For multi-label classification, a softmax activation is preferred over sigmoid activation at the output layer.
- (b) (1 point) Deep Convolutional Neural Networks reduce the spatial resolution in higher layers (closer to the output) strictly to reduce the computational complexity of the model.
- (c) (1 point) The Complete IoU loss function is effectively the same as the Generalized IoU loss function.
- (d) (1 point) For a segmentation model, an encoder-decoder model (like SegNet or U-Net) is expected to work better than a Fully Convolutional Network type architecture.
- (e) (1 point) The Feature Pyramid Network in the YOLO model trades-off finer-grained spatial features for semantic features.
- (f) (1 point) The skip connection in ResNet models enables training of deeper architectures.
- (g) (1 point) For a highly imbalanced classification dataset (Class 0: 99%, Class 1: 1%), the most appropriate evaluation metric is the Classification Accuracy.

Solution:**Rubric:**

0.5 for each correct boolean;

0.5 for correct explanation,

0 if boolean incorrect no further evaluation will be done

- (a) (1 point) **False.** In multi-label classification, where an instance can belong to multiple non-exclusive classes simultaneously, a sigmoid activation is preferred. Sigmoid treats each output node as an independent binary classification (0 or 1), whereas softmax forces the sum of all class probabilities to 1, assuming classes are mutually exclusive.
- (b) (1 point) **False.** The spatial resolution is primarily reduced to increase the receptive field of the higher layers so that they can leverage the broader spatial context to extract more meaningful, semantically rich features that assist with the downstream tasks.
- (c) (1 point) **False.** Complete IoU (CIoU) is an enhancement over Generalized IoU (GIoU). While both address non-overlapping bounding boxes, CIoU incorporates three geometric factors—overlap area, central point distance, and aspect ratio, whereas GIoU primarily focuses on the volume of the convex hull to bridge non-overlapping gaps.
- (d) (1 point) **True.** Encoder-decoder architectures (like U-Net) generally outperform basic Fully Convolutional Networks (FCN) because the symmetric decoder and skip connections help recover fine-grained spatial information and delineate boundaries more accurately.

- (e) (1 point) **False.** The Feature Pyramid Network (FPN) is specifically designed to avoid this trade-off. By fusing high-level semantic features from deeper layers with low-level, high-resolution spatial features from earlier layers, it creates a “pyramid” of feature maps that are both semantically rich and spatially detailed.
- (f) (1 point) **True.** Skip connections in ResNet mitigate the vanishing gradient problem by allowing gradients to flow directly through the identity shortcut. This enables the training of ultra-deep networks (100+ layers) that would otherwise suffer from optimization degradation.
- (g) (1 point) **False.** For highly imbalanced datasets, Classification Accuracy is misleading. A model could achieve 99% accuracy by simply predicting the majority class every time, failing to detect the critical 1% minority class. Metrics like F1-score, Precision, or Recall are much more appropriate.

2. (7 points) **Evaluation metrics [CO-4]**

- (a) (2 points) What is the difference between semantic and instance segmentation? What evaluation metric would you use to compare performance of models tackling these two tasks? How would the computation of this metric differ across these tasks?
- (b) (2 points) Write the step-by-step algorithm for computing the mean Average Precision (mAP) score. You may assume that you have predicted bounding boxes and their corresponding confidence scores and ground-truth bounding boxes.
- (c) (1 point) What are two limitations of the mAP score? How does the COCO mAP variant overcome these limitations?
- (d) (2 points) What are the different kinds of errors in object detection, as one may discover using a tool like TIDE?

Solution:

- (a) (2 points) **Semantic Segmentation:** This is the task of clustering parts of an image together which belong to the same object class. It treats multiple objects of the same category as a single entity.

Instance Segmentation: This task goes a step further by identifying and delineating each individual object of interest. It combines object detection (finding the box) with segmentation (finding the pixels).

Evaluation Metric: The evaluation metric to be used would be a mean Intersection over Union (mIoU), which takes the set of pixels for the predicted class and the corresponding set of pixels from the ground-truth to compute the IoU. For semantic segmentation, one can simply compute this for a class over the entire dataset, whereas for instance segmentation, this needs to be done for each instance present in each image (similar to mAP) and then aggregated over the dataset.

Rubric:

0.5 for Semantic Segmentation;
0.5 for Instance Segmentation.
0.5 for What Evaluation metrics
0.5 for how Evaluation metrics differs

- (b) (2 points) Following are the steps for computing the mAP score:

1. **Preprocessing and IoU Calculation.** For a specific class, collect all predicted bounding boxes across the entire test set and rank them by their confidence score (highest to lowest).

For every predicted box, calculate the Intersection over Union (IoU) with all ground-truth boxes of the same class in that specific image.

2. **Matching Predictions to Ground Truth.** Iterate through the ranked predictions. A prediction is labeled a True Positive (TP) or a False Positive (FP) based on a predefined IoU threshold (e.g., $\text{IoU} \geq 0.5$):
 - True Positive (TP): The IoU exceeds the threshold and the ground-truth box has not been "claimed" by a higher-confidence prediction.
 - False Positive (FP): The IoU is below the threshold, OR the ground-truth box has already been assigned to another prediction.
 - False Negative (FN): A ground-truth box that was never matched to any prediction.
3. **Calculate Precision and Recall.** Create a running tally of TPs and FPs as you go down the ranked list.
 - Precision = $\frac{TP}{TP+FP}$
 - Recall = $\frac{TP}{TP+FN}$
4. **Compute Average Precision (AP) for one Class.** The AP is the area under the Precision-Recall (PR) curve. This could be computed with or without interpolation for additional smoothness.
5. **Calculate the Mean (mAP).** Finally, average the AP scores across all object classes in the dataset.

Rubric:

0.5 for Step 1;
 0.5 for Step 2.
 1 for Step 3,4,5

(c) (1 point) The two limitations of mAP are:

1. Sensitivity to the IoU Threshold.
2. Lack of Granularity regarding Object Size.

The COCO Solution: (1) COCO calculates $\text{mAP}@[0.5 : 0.05 : 0.95]$. This is the average AP computed at 10 different IoU thresholds, from 0.5 to 0.95 with a step size of 0.05. (2) COCO introduced scale-specific metrics. It reports mAP separately for different object size categories: Small, Medium and Large.

Rubric:

0.25 for each limitation;
 0.5 for COCO mAP.

(d) (2 points) The different types of errors that can be analyzed by a tool like TIDE are:

1. **Classification Error (Cls):** The model correctly localized the object (high IoU with ground truth) but assigned it the wrong class label.
2. **Localization Error (Loc):** The model identified the correct class, but the bounding box is imprecise (the IoU is below the success threshold, typically 0.5, but above a minimal background threshold).
3. **Both Cls and Loc Error:** The model is both misclassified and poorly localized. The prediction has some overlap with a ground truth object of a different class, but not enough to meet the success threshold.
4. **Duplicate Detection Error (Dupe):** The model predicted multiple correct boxes for the same single ground truth object. Only the highest-scoring detection is counted as a "True Positive" while others become "False Positives".

5. **Background Error (Bkg):** The model “hallucinates” an object where none exists. The predicted box has very little to no overlap with any ground truth object.
6. **Missed GT Error (Missed):** A ground truth object that the model failed to detect entirely. These are strictly “False Negatives”.

Rubric:

0.5 for each limitation with explanation;
0.25 for each limitation without explanation.

3. (6 points) **Camera Geometry [CO-2]**

Recall that the intrinsic (internal) camera matrix comprises of the parameters relevant to the internal camera components, while the extrinsic (external) matrix comprises of the parameters relevant to the world and camera coordinate systems.

- (a) (2 points) How many degrees of freedom does the intrinsic camera matrix have? What are the different parameters that comprise these degrees of freedom?
- (b) (2 points) A camera is first rotated about its Y-axis by $\frac{\pi}{2}$, followed by a rotation about its original X-axis by $\frac{\pi}{2}$. Find the composite rotation matrix $\mathbf{R}$. How would you find the axis of the combined rotation using $\mathbf{R}$? How is this axis related to the eigenvectors of $\mathbf{R}$?
- (c) (2 points) A camera was moved via a pure rotation (i.e., $\mathbf{R}_{3 \times 3}$, $\mathbf{T}_{3 \times 1} = 0$) to capture two images of the same scene (with overlapping Fields of View). Show that the images (i.e., their pixels) are related by a 3×3 transformation matrix $\mathbf{H}$, which can be written in terms of the camera's intrinsic matrix ($\mathbf{K}_{3 \times 3}$) and extrinsic parameters ($\mathbf{R}, \mathbf{T}$). Write the matrix $\mathbf{H}$ using $\mathbf{K}$, $\mathbf{R}$, $\mathbf{T}$ as needed. {Hint: Use the image formation pipeline.}

Solution:

- (a) (2 points) The intrinsic camera matrix, typically denoted as $\mathbf{K}$, characterizes the internal parameters of a camera that effectively maps coordinates from the 3D camera frame to the 2D image plane. The intrinsic camera matrix has *5 degrees of freedom*:

Rubric:

0.5 for each correct parameter of matrix; 4 needed at least

1. f_x : Focal length as expressed in terms of horizontal length of pixels.
 2. f_y : Focal length as expressed in terms of vertical length of pixels.
 3. c_x : Horizontal coordinate of the principal point, i.e., the projection of the camera centre on the image plane.
 4. c_y : Vertical coordinate of the principal point, i.e., the projection of the camera centre on the image plane.
 5. s : The Skew coefficient, which accounts for any non-orthogonality (slant) between the x and y axes of the pixel grid. In most modern sensors, this is effectively 0.
- (b) (2 points) **1. Individual Rotation Matrices.** For a right-handed coordinate system, the rotation matrices for a rotation $\theta = \frac{\pi}{2}$ about the Y and X axes are defined as:

Rubric:

0.5, 0.5 for each correct rotation matrix;
0.5 for correct composition,
0.5 for correct normalized vector or relation of eigenvector

$$\mathbf{R}_y\left(\frac{\pi}{2}\right) = \begin{bmatrix} \cos\left(\frac{\pi}{2}\right) & 0 & \sin\left(\frac{\pi}{2}\right) \\ 0 & 1 & 0 \\ -\sin\left(\frac{\pi}{2}\right) & 0 & \cos\left(\frac{\pi}{2}\right) \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix} \quad (1)$$

$$\mathbf{R}_x\left(\frac{\pi}{2}\right) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\frac{\pi}{2}) & -\sin(\frac{\pi}{2}) \\ 0 & \sin(\frac{\pi}{2}) & \cos(\frac{\pi}{2}) \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \quad (2)$$

2. Composite Rotation Matrix $\mathbf{R}$ Since the rotations are performed relative to the original, fixed axes (extrinsic rotations), the composite rotation matrix is computed by multiplying the matrices in reverse order of application:

$$\mathbf{R} = \mathbf{R}_x\left(\frac{\pi}{2}\right) \mathbf{R}_y\left(\frac{\pi}{2}\right) \quad (3)$$

$$\mathbf{R} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad (4)$$

3. Finding the Axis of Rotation The axis of rotation $\mathbf{v} = [v_x, v_y, v_z]^T$ can be extracted from the antisymmetric part of $\mathbf{R}$. The vector components are proportional to:

$$v_x = R_{32} - R_{23} = 1 - 0 = 1$$

$$v_y = R_{13} - R_{31} = 1 - 0 = 1$$

$$v_z = R_{21} - R_{12} = 1 - 0 = 1$$

The normalized unit vector $\hat{\mathbf{v}}$ is:

$$\hat{\mathbf{v}} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

4. Relationship to Eigenvectors

The axis of rotation is the **eigenvector** of $\mathbf{R}$ corresponding to the eigenvalue $\lambda = 1$.

Justification: Any vector pointing along the axis of rotation remains unchanged by the transformation. This is defined by the eigenvalue equation:

$$\mathbf{R}\mathbf{v} = 1 \cdot \mathbf{v}$$

By Euler's Rotation Theorem, every proper rotation matrix in 3D space has an eigenvalue $\lambda = 1$. The associated eigenvector defines the stationary line, which is the physical axis of rotation.

- (c) (2 points) To show that two images captured by a camera undergoing pure rotation are related by a 3×3 transformation matrix $\mathbf{H}$, we analyze the projection of a 3D point $\mathbf{P}$ into the two image planes.

Rubric:

0.5, 0.5 for each correct projection equation;

0.5 for removing the translation,

0.5 for deriving the correct $\mathbf{H}$ (homography) matrix

1. Camera Projection Model. A 3D point $\mathbf{X}_c$ in the camera's local coordinate frame is projected to pixel coordinates $\mathbf{x} = [u, v, 1]^T$ via the intrinsic matrix $\mathbf{K}$:

$$\mathbf{x} \simeq \mathbf{K}\mathbf{X}_c \implies \mathbf{X}_c \simeq \mathbf{K}^{-1}\mathbf{x} \quad (5)$$

where $\simeq$ denotes equality up to a scale factor.

2. Relationship Between Camera Frames. Let the first camera be the reference frame. The coordinates of the 3D point in the first frame are $\mathbf{X}_{c1}$. The second camera is related to the first by a rotation $\mathbf{R}$ and a translation $\mathbf{T}$. The coordinates in the second frame $\mathbf{X}_{c2}$ are:

$$\mathbf{X}_{c2} = \mathbf{R}\mathbf{X}_{c1} + \mathbf{T} \quad (6)$$

For a **pure rotation**, the translation vector is zero ($\mathbf{T} = \mathbf{0}$), simplifying the expression to:

$$\mathbf{X}_{c2} = \mathbf{R}\mathbf{X}_{c1} \quad (7)$$

3. Pixel-to-Pixel Transformation. Substituting the pixel-to-3D relationship from Equation (1) into the rotation relationship in Equation (3):

$$\mathbf{K}^{-1}\mathbf{x}_2 \simeq \mathbf{R}(\mathbf{K}^{-1}\mathbf{x}_1) \quad (8)$$

To isolate the second image coordinates $\mathbf{x}_2$, we multiply both sides by $\mathbf{K}$:

$$\mathbf{x}_2 \simeq (\mathbf{K}\mathbf{R}\mathbf{K}^{-1})\mathbf{x}_1 \quad (9)$$

4. The Transformation Matrix $\mathbf{H}$. The images are related by the 3×3 transformation matrix $\mathbf{H}$ (a homography), such that $\mathbf{x}_2 \simeq \mathbf{H}\mathbf{x}_1$. Based on the derivation above, the matrix is defined as:

$$\mathbf{H} = \mathbf{K}\mathbf{R}\mathbf{K}^{-1} \quad (10)$$

Under the condition of pure rotation ($\mathbf{T} = \mathbf{0}$), the transformation $\mathbf{H}$ is independent of the depth of the 3D points. This allows for image stitching and panorama construction without explicit 3D reconstruction of the scene (as we will see later in the course).